Predictir

Execution research and database audit

November 4, 2026 | America/New_York

This supplement records the pre-release verification of the execution model and database changes. Production rollout and subsequent checks are documented separately.

The new model studies execution cost, quote dynamics and inventory risk. It preserves the independent event-outcome forecast. Established research motivates each component; this report documents the actual implementation, measured experiments and remaining empirical gaps.

Evidence	What the measurement establishes
32 category cases	Synthetic adapters cover eight categories, both sides and both actions; they do not validate category-specific forecasts.
Five toggle ablations	Each switch changes its named numerical computation. Unchanged independent settlement probabilities are checked.
Quote-signal controls	Planted and null synthetic processes test the local fit/holdout gates; they are not market returns.
Stock simulations	Three separate 10,000-path liquidation scenarios compare execution cost/risk using common random paths; no profitable stock strategy is claimed.
Local CPU benchmark	512 synthetic quote events: p50 1.256 ms; p95 1.378 ms over 41 measured trials. Excludes network, browser and exchange time.

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The accompanying research audit covers the broader market workbench. This supplement focuses on equations, experiments, implementation checks, and database findings.

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Inspectable switch comparisons

These ablations use a seeded synthetic process with a planted quote signal and arrival clustering. They demonstrate working controls under stated conditions. A live contract without sufficient aligned observations correctly withholds those fitted adjustments.

Switch	Measured output	On	Off
order_flow	quote_reference_change	-0.00028280388	0
arrival_clustering	arrival_risk_multiplier	1.7320508	1
inventory_skew	inventory_reference_change	-0.06	0
optimal_schedule	first_slice_change	5.613778	0
enabled	first_slice_change	5.613778	0

All switches preserve the supplied independent settlement probability. A failed quote-model gate does not force a different forecast merely to make a toggle appear active. The schedule's objective includes a chosen risk preference; an objective reduction is not an expected profit or a latency improvement.

Implementation and verification

This records the verification completed on November 4, 2026 before the initial production release. See the separate production reliability verification for the rollout and subsequent checks.

Project presentation

GitHub About and the README describe Predictir as prediction market research and simulated trading. The README links to the public app and summarizes its architecture. The presentation update reached main in `c2aea8ae2130ac44aa7c0dc0804796730e420b90` while the application build was held for review.

Database correctness

All three primary keys were verified against the live catalog using read-only catalog/statistics queries. No production writes, index creation, schema migration, grants, vacuum, truncation or destructive maintenance was performed. The suggested quota-overrun defect did not apply to the actual validated `limit-amount` binding.

Implemented duplicate record-write suppression, database-authoritative quota admission, post-lock lease deadlines, stale-producer fencing, atomic X-cursor publication and safe recovery from local refresh contention or temporary X budget denial. Existing sliding expiry and immutable journal rules were preserved. An additional `(pk)` index was not justified by measured read/write/storage tradeoffs. Both positive and negative A/B results remain in the report.

Database verification passed 302 tests with zero failures and two opt-in embedding and vector tests skipped. Type checking passed. PostgreSQL integration tests used a disposable local `query_runtime_test` database. The experiment used a separate local `query_optimization_test` database, and its benchmark schemas were removed after verification.

Execution model and research tools

The selected-contract interface now has a fifth tab, Execution lab. The existing MCP server already had five tools; `analyze_execution_lab` is the sixth. The assistant receives the same validated controls and selected forecast model, scoped to the exact ticker. A standard explanation can read back deterministic tool figures without a language-model generation charge. Other assistant questions retain bounded tool calls and existing allowance controls.

The model composes observed order-flow imbalance and a chronological next-quote challenger, exponential Hawkes quote-arrival likelihood, inventory-sensitive quote scenarios, a discrete tridiagonal execution optimizer and finite hold-versus-exit utility. Switches affect these calculations. They do not rewrite the independent settlement probability. The quote challenger and arrival model can remain unapplied when their local gates fail; unavailable features are not replaced with invented data.

The native feed retains a bounded, per-contract circular history of validated best queues before browser throttling. It explicitly requests non-conflated updates, checks exchange subscription sequence continuity, and resets the research generation on snapshots, gaps, disconnects or a one-sided book. The lab captures book and event history together. Unaligned or stale data cannot adjust another book. The arrival model decays through observed silence only when the connection establishes that observation window. Quote updates are not mislabeled as trades or news.

Independent review found and resolved book/window mismatch, a Hawkes training cutoff that depended on the first holdout timestamp, missing idle-time decay, a derived-volatility boundary failure, dimensional labels and unverified fee assumptions. Dedicated counterexamples verify the fixes. Unknown fee metadata retains an explicit default scenario and withholds a released exit decision. Fill estimates retain the same two-cent slippage convention and per-level cent rounding as existing paper execution.

Evidence, limits and reproduction

The final combined suite passed **331 tests, zero failures**, including actual local PostgreSQL integration and 13 TypeScript tests. Two opt-in tests for the pinned embedding model and native vector/RLS path were intentionally skipped in this combined run; their earlier separate checks remain documented in V3 verification. The production build and type checking passed. No pre-commit configuration exists.

The research runner records 32 synthetic category/side/action cases, five numerical toggle ablations, five planted-signal controls and five independent-noise controls. Separate stock experiments use three sets of 10,000 paired liquidation paths, including jump stress and a historical IBM daily-volatility scale. They measure conditional execution cost and variance, not stock or Kalshi strategy profitability. Short holdouts, repeated rolling tests, assumed impact parameters and the daily-to-intraday conversion are documented rather than hidden.

The source ledger tracks all 97 requested domain/title entries: 78 exact targets opened, six resolved targets blocked and 13 unresolved or ambiguous. Reading scopes distinguish abstracts, selected sections and metadata checks. Twenty additional primary papers/documents were checked; three existing project references are disclosed, leaving 17 canonical URLs new to the scanned documentation. A failed open does not receive a successful-read checkbox.

The source ledger addresses inventory-price intuition, GLFT signs and solution methods, OFI data requirements, stock and binary modeling assumptions, DeepLOB training and inference, cache lines, kernel isolation, and the distinction between LMSR and an order book. The experiments measure the implemented numerical models and data paths.

The UI was exercised with an actual public Golden State contract and current REST book. At the observed snapshot, 20 YES contracts required $13.52 including $0.32 modeled fees. Raising the supplied risk preference changed the first target slice from about 1.715 to 11.111 contracts and reduced assumed timing variance while increasing expected execution cost. Disabling the model restored equal 1.667-contract slices and the same independent P(YES), about 57.3%. These are dated UI observations, not expected future prices or returns. No native streaming key was configured in the local preview, so live fitted features correctly displayed missing continuous history. The native-window integration was verified with deterministic stream fixtures and independent counterexamples.

The new tab supports numeric controls, keyboard tab navigation, bounded inputs, stale-result labeling, cancellation, ticker/model changes and JSON export. The connected browser-to-backend-to-MCP explanation reproduced the current 20-contract costs and controls without a language-model generation. The browser's export produced a parseable scenario JSON file with its dated observations and assumptions. All eight category adapters render in tests. Desktop browser review and responsive CSS checks do not establish a native mobile-device visual test; no such result is claimed.

Run the documented tests from `web/` and use `node tools/validate_execution_research.mjs` for the seeded experiments. The PDF builder uses pinned ReportLab dependencies. Report inputs include public source observations, local benchmark results, and synthetic fixtures.

Execution model, equations and validation

Predictir's Execution lab adds a separate, selectable research layer for the cost and risk of a hypothetical paper order. It combines observed best-queue updates, a next-quote challenger, quote-arrival clustering, inventory-aware quote scenarios and a discrete execution schedule. Every feature has a real calculation and an off state. It never replaces an independent event-outcome probability with market price, order-flow pressure, an assumed diffusion or news popularity.

The fifth contract-detail tab exposes this layer. Existing MCP tools remain intact; the new execution research tool is the sixth registered MCP tool. The backend, assistant and MCP use the same normalized controls and calculation. The application changes remain pending deployment; this document does not establish that the public production website runs this branch.

Inputs and provenance

`web/lib/workspace/execution-research.mjs` exports `normalizeExecutionControls`, `almgrenChrissSchedule` and `executionResearch`. `web/lib/workspace/microstructure-model.mjs` contains the separate next-quote model and quote-arrival model. Numeric controls reject strings, booleans, non-finite values, unknown keys and oversized requests. The control limits bound fitting and schedule work: 10,000 contracts, 60 intervals, a horizon of 30-86,400 seconds, and at most 2,048 input quote events. The retained contiguous history is capped at 1,024 events. Observed best queues above one trillion contracts are rejected to avoid nonsensical/overflowing derived metrics. Raw volatility remains bounded at one; the analytic schedule accepts the full possible derived value up to `sqrt(3)` after the capped arrival multiplier, without silently discarding the selected risk scenario.

The backend supplies the exact contract's current, fresh, two-sided fixed-point book, its existing independent forecast packet, the verified series fee coefficient, and a bounded window from the existing exchange stream. It does not open another streaming connection for this feature. YES asks are derived from complementary NO bids and vice versa. Immediate paper-entry and exit estimates use the existing `executionCurve`, including a two-cent slippage limit, depth limits and per-level upward cent rounding. A partial bid fill withholds a full-position exit value. These executable snapshot estimates are separate from future schedule-cost scenarios.

Quote history contains monotonic local receive times, ticker-local sequence numbers, a snapshot-generation identifier and observed best prices/queues. It resets across missing, reordered, malformed or new-generation events. Equal receive timestamps preserve ordered OFI but withhold Hawkes fitting rather than inventing microseconds. Sequence continuity and a non-conflated stream request do not prove that the exchange exposed every underlying order action. The system cannot recover hidden orders, an individual order's queue position, or cancellation intent from net queue updates. Stale observed history is excluded by the backend wrapper.

Volatility, temporary impact and risk aversion are editable, unit-labeled assumptions. A single live snapshot does not calibrate them. The inventory quote scenario is bounded to [0,1] and rounded outward to cents. Its observed-spread anchor is not a fitted arrival-versus-distance function, so the result is not advertised as the full Avellaneda-Stoikov or GLFT optimum.

Separate next-quote model

For observed best prices `B,A` and sizes `qB,qA`, an OFI transition is

```
e = 1[Bnew >= Bold] qBnew - 1[Bnew <= Bold] qBold - 1[Anew <= Aold] qAnew + 1[Anew >= Aold] qAold.
```

The original empirical OFI paper uses Level I best queues; full Level II/III data are not necessary for this definition. Multi-level OFI is a different extension. The paper demonstrates a short-interval relationship in its equity sample, not universal predictive alpha or a Kalshi settlement probability. Cont, Kukanov and Stoikov, sections 2.1-2.2.

The challenger predicts the **next observed quote midpoint change**, using current OFI divided by mean best depth and current queue imbalance. The target is strictly later than the feature observation. It uses two-feature ridge regression with a fixed penalty and training-only normalization. At least 152 contiguous events are required. A chronological 60/20/20 split fits coefficients, calibrates residual quantiles and evaluates an untouched holdout. It compares against zero midpoint change, not a hindsight-selected alternative.

The local application gate requires a strictly positive lower endpoint of a seeded, paired, moving-block bootstrap interval for squared-error improvement and at least 70% held-out coverage for the calibrated nominal 80% residual interval. The bootstrap uses 300 replicates, five-event circular blocks and a fixed seed. It does not refit on the holdout. A failed gate applies zero shift. Even a passed gate changes only the short-horizon quote reference; the independent settlement probability is copied unchanged. A rolling short sample, overlapping windows and repeated testing do not establish prospective significance, realized profits or stable alpha. The interval does not include all model uncertainty.

Arrival clustering

The second model fits an exponential Hawkes intensity `lambda(t)=mu + alpha sum exp(-beta(t-ti))` to **observed quote-update arrivals**. The amplitude `alpha` and decay `beta` are both measured per second; the branching ratio is `alpha/beta`. It is constrained below one, with fitting capped at 0.95. This convention differs from sources that parameterize the kernel as `branching_ratio * beta * exp(-beta*t)`.

The likelihood uses an O(n) decaying-memory recursion and an analytic compensator, including the history before a held-out window. Fitting uses a bounded beta grid and constrained EM updates with multiple starts; it is not a claim of a global maximum. A chronological 70/30 comparison requires both a training AIC advantage over Poisson and positive held-out log-likelihood gain. The training exposure ends exactly at its last observed training event; no held-out timestamp sets that boundary. The held-out exposure ends at its last observed event, without inventing a future no-arrival interval. Changing the first held-out timestamp cannot change fitted parameters or the training Poisson rate. The same past conditions both held-out likelihoods. Fewer than 120 events, tied receive timestamps or failed gates leave the execution-risk multiplier at one. Bacry, Mastromatteo and Muzy, exponential-kernel example and Appendix C.1.

For an aligned, continuously connected stream window, the backend supplies its actual observed end time; the compensator and intensity decay through observed idle time after the final quote. A disconnected interval is not treated as evidence that no events occurred. Offline calls without that end time label intensity at the last observed quote instead of claiming it is current. The fit boundary and parameters remain independent of the later idle interval.

When enabled and locally accepted, the bounded multiplier is `sqrt(clamp(current_intensity / training_Poisson_rate, 0.5, 3))`. Applying this to assumed volatility is an explicit **scenario assumption**, not an empirically fitted volatility relationship. A fitted cluster process is not a direction, toxicity or event-outcome model. Nonstationary bursts and non-exponential kernels can violate this simple model.

Execution and finite settlement decisions

For a requested quantity `Q`, horizon `T` in minutes, `N` intervals, temporary impact `eta`, volatility `sigma` and risk preference `lambda`, the schedule minimizes

```
J = fixed_cost + eta/dt * sum(nk^2) + lambda * sigma^2 * dt * sum(xk^2)
```

subject to `x0=Q`, `xN=0`, and `nk=x(k-1)-xk`. Here `sigma` is per-contract price volatility in USD per contract per square-root minute; `eta` has units USD * minute/contracts^2, so `eta * trading_rate` is per-contract price impact. The tridiagonal system has diagonal `2 + lambda*sigma^2*dt^2/eta` and off-diagonal `-1`; a Thomas solve takes O(N) time and memory. It avoids overflow in hyperbolic functions for extreme allowed parameters. Zero risk or disabling optimal scheduling yields equal-quantity TWAP exactly. Trades occur at the **start** of each displayed interval; `remaining` is inventory exposed during that interval. This convention matches the reported conditional variance. The last child is at `T-dt`, with inventory zero during the last interval.

The objective includes a fixed current-spread and unrounded reference-fee estimate. Current depth-based entry/exit quotes use actual per-level rounding separately. Fractional schedule quantities are target allocations, not orders or fill promises. Constant impact, zero drift, no permanent impact, known variance and an unchanged future liquidity regime are assumptions. The discrete model's optimum can have higher expected execution cost than TWAP while reducing its risk penalty enough to lower `J`. A lower objective is not profit. The continuous `sqrt(lambda*sigma^2/eta)` decay formula is a limit; finite time steps have a discrete correction. Almgren and Chriss, sections 1.5-2.2.

For a hypothetical existing binary position of quantity `q`, the hold-versus-sell calculation compares current full executable net bid proceeds against the CARA certainty equivalent

```
CE(p,q,risk) = -log((1-p) + p*exp(-risk*q))/risk,
```

with the risk-neutral limit `p*q`. The implementation uses stable `log1p`/`expm1` arithmetic and handles probabilities exactly zero or one. Entry cost is sunk in this comparison. Lower and upper model probabilities give a range of certainty equivalents. A missing, stale, mismatched or unapproved forecast withholds a released position action. An unapproved forecast can show a separately named experimental `scenario_decision`; it does not become a released recommendation. No optional-stopping theorem turns a negative expected value after fees into an edge. The paper ledger remains responsible for actual depth-based fills and settlement.

Attachment gap analysis that changed implementation

Topic	Correction and resulting choice
Long inventory	Lower bid/ask quotes discourage additional sellers and encourage buyers to take inventory. The attached paper reverses those counterparties. Inventory quote direction is explicitly tested.
HJB and linear methods	A maximum makes a general HJB nonlinear, but GLFT transforms its particular problem into linear ODEs and spectral approximations. A blanket assertion that linear methods fail is too strong. The execution layer solves its own finite convex problem directly. GLFT.
Reservation price	It is a subjective inventory valuation under assumptions, not a guaranteed tradable fair price or a settlement forecast. The lab labels it a quote scenario. Avellaneda-Stoikov, sections 2.2, 3.
OFI inputs	Best-queue event changes suffice for classic OFI; occasional snapshots cannot recover omitted event order. The lab uses only its latest observed contiguous stream generation and gates missing history.
Hawkes inference	Self-exciting activity does not determine direction or remove bots. Only quote-arrival timing is fitted; no invented sentiment scores or accuracy histories enter the probability.
A-C precision	Finite-step corrections, units and the cost-versus-risk distinction matter. The implementation solves the actual discrete objective rather than treating its continuous limit as exact.
LMSR	LMSR is an alternative automated-market-maker mechanism. Kalshi's displayed order book is a CLOB; LMSR cost/softmax equations cannot replace its depth or fees. No LMSR fills are represented as Kalshi fills.
Deep learning	A CNN/LSTM architecture requires a suitable event-history dataset, leakage controls, calibration and fee-aware tests. Its name or greater parameter count does not substantiate better event probabilities. No untrained DeepLOB model is presented as a feature. Deep LOB forecasting survey.
Hardware latency	NIC-specific kernel bypass, cache-line padding and busy polling depend on hardware, scheduler, runtime and workload. They do not confer nanosecond guarantees on a hosted Node application. Existing engineering benchmarks remain separate.

Topic	Correction and resulting choice
Synthetic results	Positive controls, simulations, local CPU measurements and model-conditional objective changes are named separately from empirical market predictions and realized results.

Validation and reproduction

Run `node --test tests/execution-research.test.mjs` from `web/`. Tests compare OFI to independent arithmetic, Hawkes recursion to a quadratic likelihood oracle, and the execution optimum to a coarse exhaustive search. They cover quantity conservation, zero-risk TWAP, extreme parameters, future-holdout mutation, both market sides, all eight categories, existing per-level fee calculations, stale books, missing forecasts, partial depth and strict input validation. The core tests make no external requests and use no credentials.

Run `node tools/validate_execution_research.mjs` for the bounded research experiment. The inspectable output is `review/execution-research-validation.json`, with source hashes and runtime details. Its 32 category/side/action cases use explicitly synthetic profiles to test adapters and mathematical invariants. Five toggle ablations each change an actual numeric result while preserving settlement probabilities. Five planted-signal controls pass the local gate; five independent-noise controls do not. Those outcomes check the validation mechanism, not real-market accuracy. The CPU benchmark retains all 41 measured observations after five warmups; p50/p95/p99 are local empirical calculation percentiles with thin tail resolution, excluding network, storage, rendering and exchange latency.

The runner also performs three separate 10,000-path stock liquidation simulations using the same random increments for the model and TWAP: an assumed zero-drift diffusion, a symmetric jump stress, and a volatility scale estimated from historical IBM daily closes. Optional `--fetch-stock` makes one bounded, unpaid request to Alpha Vantage's documented IBM demonstration endpoint. The response is cached under ignored `web/.research/`; raw provider data are not committed or publicly redistributed. The report records response hash, provider, retrieval time and date bounds. Subsequent runs use the local cache. A clean installation without that private cache still runs the synthetic diffusion and jump experiments; the historical-scale case is explicitly absent until fetched.

The observed IBM input had 100 daily closes from June 15 to November 4, 2026. The daily return split was 59 fit, 20 calibration and 20 untouched holdout observations. A zero-return forecast's calibration 90% interval covered 85% of that small holdout. This is a descriptive **single-asset return-interval check**, not probability calibration, model selection across stocks or evidence of profitable prediction. The fitted daily return RMS is converted to a per-minute dollar volatility using `reference_price * RMS / sqrt(390)`, an assumption of uniform intraday variance that ignores overnight/intraday differences. Survivorship, short history, corporate actions, regimes and provider limitations remain. No intraday order-flow or impact dataset was inferred from daily closes. Alpha Vantage daily-series documentation.

In the assumed 0.20 USD/sqrt-minute diffusion example, the model trades faster than TWAP: analytic expected cost is about $66.19 versus $22.67, while conditional variance is about 12,376 versus 1,520,889 USD^2. With the chosen 0.001 inverse-USD risk preference, the objective is $78.56 versus $1,543.56. Paired Monte Carlo results approximate those conditional moments; adding jump stress

increases variance and demonstrates model sensitivity. These are simulated costs for a 2,000-share liquidation under supplied impact/spread assumptions. They are neither dollar profits nor an observed benefit to a real trader.

No production migration, new paid provider, extra stream connection or cloud deployment is required by this math layer. Model promotion still requires a frozen prospective protocol, sufficient independent event data and fee-aware executable outcomes.

Database correctness and production evidence

The changes address duplicate writes, incorrect local quota denials and stale cache producers. They add no database index, schema migration, service, connection pool or paid resource. Production was inspected read-only; no production SQL writes, DDL, code push or deployment occurred. The backend remains on PostgreSQL 17 with the existing four-query-connection pool and separate LISTEN connection.

Evidence and measurement window

The sanitized production snapshot in `review/query-production-aggregate-snapshot.json` was observed at **2026-11-05 02:37:22.895 UTC**. Server PostgreSQL 17.11 started at **2026-11-04 22:58:48.156 UTC**. The original inspection did not read `pg_stat_statements_info.stats_reset`; statement counters are cumulative since their collector reset, whose exact time is unavailable. Server startup is a separate timestamp and does not establish the statistics window. These aggregates provide mean, standard deviation, minimum and maximum, not p95/p99.

The snapshot estimated 478 live `records` rows, 2,269 updates and zero HOT updates. Physical storage was 294,912 bytes of heap, 131,072 bytes of table indexes and 58,261,504 bytes of TOAST relation storage including its indexes. Quotas recorded 5,270 updates, all HOT. The dominant record UPSERT had 2,735 calls, mean server execution 5.100 ms, maximum 394.404 ms, standard deviation 13.846 ms and 97,193,608 cumulative WAL bytes. It read one shared block and otherwise mostly hit shared buffers. No active lock wait was observed in that inspection. This supports investigating large JSON/TOAST update work; physical TOAST size alone does not prove bloat, and an empty wait snapshot does not prove contention never occurs.

The paired synthetic A/B experiment is separately documented in query-optimization-benchmark.md, with all observations and sanitized plans in `review/query-optimization-benchmark.json`. It used isolated local databases, 256 and 100,000 rows, four payload profiles, five warmups and 41 measured trials per variant. Its measured p50/p95 values describe those local trials, not production percentiles or projected savings.

Implemented changes

Exact duplicate record writes: `save(item,true)` retains the existing UPSERT behavior but updates only when JSONB payload or expiry differs. JSON key ordering alone does not create a new tuple. Changed payloads and TTL changes are still written; strict `put` still reports a duplicate as a conflict. In the 100,000-row / 32-KiB high-entropy duplicate fixture, baseline p50/p95 was 0.602/0.683 ms with about 36,448 WAL bytes per attempt; the plain guard measured 0.119/0.145 ms and 54 bytes. The guard still locks the conflicting row and can emit lock WAL. It is not a zero-I/O or zero-contention operation.

The measured equal-payload CASE can preserve a TOAST pointer when only the SQL expiry changes, but it is **not adopted** in this stage. Existing payloads embed `expires_epoch`, and cache values often include a new retrieval timestamp, so real refreshes can change the JSON as well. CASE also increased CPU cost for changed payloads in the experiment. No TTL normalization or frozen-journal format change was introduced.

Database-authoritative quota decisions: the key-only 60-second denial map was removed. It could incorrectly deny a smaller admissible amount, a raised limit, an expired window, an external refund or capacity restored by rollback. A local cache cannot reliably observe distributed refunds and transaction rollback. Each `consume` now makes its existing atomic SQL decision. The validated first insert still requires `amount <= limit`; the existing conflict guard still receives `limit-amount`. There was no missing quota-overrun fix.

Accepted consumes deliberately retain their existing sliding-expiry semantics. For date-bearing daily/monthly keys, the date identifies the accounting period while the TTL retains the row afterward. This change does not reinterpret all TTLs as fixed windows. Repeated denials now cost database requests: the local four-worker benchmark measured hot-key round-trip p95 5.052 ms versus 2.850 ms across four distinct exhausted keys, with no admitted requests. Existing edge admission limits, application quotas and the four-connection pool remain in place; no extra metadata query or denial-cache invalidation service was added.

Actual-clock deadlines: record freshness, quota reset decisions and lease expiry use `clock_timestamp()` where wall-clock expiry is required. A conflicting lease assigns its new deadline in the post-lock UPDATE rather than reusing `EXCLUDED.expires_at`, which was computed before waiting. A controlled 350-ms lock wait with a 200-ms lease verifies that the acquired lease still has usable lifetime afterward. Long-transaction tests separately verify record expiry, quota reset and new lease lifetime against actual time. PostgreSQL time semantics.

Fenced cache publication: after acquiring a refresh lease, `cached` checks the cache again before charging upstream work. Cache publication locks the lease row in one materialized step, then checks its current owner and wall-clock expiry in a separate step before writing. Both the final cache record and optional producer-side records use this fence. An expired producer cannot overwrite a fresh successor, and an old release cannot delete the successor's lease. If a successor already published, the older caller receives that retained value; otherwise a typed local-refresh error is returned. The producer callback can optionally use `publishRecord` for side records such as the X cursor; existing callbacks can ignore this additive context.

Local `CACHE_LEASE_LOST`/`CACHE_LEASE_BUSY` coordination errors are distinct from upstream provider throttling. The social integration preserves the newest fenced cursor and does not set a shared provider cooldown merely because another process owns the refresh. Failed X budget reservations are not cached as a successful 30-day account lookup. Quota reservations still charge actual upstream work; fencing does not refund requests that were already made. Existing assistant reservation/refund transactions remain atomic and fail closed.

Index decision

The existing `records_pkey(pk,sk)` supports leading `pk` lookups as well as the exact key reads used by `get`. A separate `pk` index is narrower but duplicates storage and adds write maintenance. At 100,000 synthetic rows it added 7,839,744 bytes and gave no consistent benefit across exact, prefix and ordered journal queries. No new index, fillfactor adjustment, autovacuum setting or feature flag is justified by the reported ten-percent planner estimate alone. PostgreSQL multicolumn indexes.

Validation and reproduction

Ten actual local query regressions cover no-op tuple identity, changed JSON/expiry writes, weighted concurrent quota admission, varying request amounts/limits, expiry/refunds/rollback, sliding TTL, post-lock lease deadlines, stale-owner release, stale cache/side-record publication, typed refresh contention, delayed miss rechecks and long transactions. Five of the initial six cases failed against the baseline; sliding TTL already passed and was preserved. The revised cases pass. Existing store, assistant reservation/refund and commit-notification integration tests also pass.

Root's final combined verification, including typed busy-error coverage, used the separate disposable `query_runtime_test` database: **292 JavaScript tests passed with two intentional opt-in skips, plus 10 TypeScript tests passed; 302 total passed, zero failed.** Type checking also passed. The model/vector opt-in tests are outside this SQL change.

To reproduce local tests, supply only a loopback disposable `MARKET_TEST_DATABASE_URL` ending in `_test`, then run `node --test tests/postgres-queries.test.mjs tests/postgres.test.mjs tests/assistant-budget.test.mjs tests/runtime-database.test.mjs` from `web/`. Never point integration tests or EXPLAIN ANALYZE writes at production. The benchmark CLI additionally rejects the in-use `engineering_test` database and cleans up only its own isolated schema.

For approved read-only operational inspection, supply backend-only `DATABASE_URL` and the appropriate `DATABASE_CA_CERT` privately, then run `node tools/inspect_postgres_queries.mjs` from `web/`. The inspector uses `BEGIN READ ONLY`, bounded statement/lock timeouts, fixed catalog/aggregate queries and optional statistics-reset inspection. It excludes application rows, payloads, SQL text, bind values and credentials. Do not paste private connection settings into reports or shell history. Availability of aggregate statistics depends on existing database privileges; the tool does not grant privileges or change the database.

The inspector now checks existing schema `USAGE` and statistics-view `SELECT` privileges through PostgreSQL catalogs before each optional statistics read. Restricted roles report `insufficient_existing_privileges` without deliberately triggering SQLSTATE 42501. The reset view is checked separately; permission revocation races still roll back to a read-only savepoint. Use an already authorized monitoring connection when aggregate statistics are needed, rather than granting monitoring privileges to the application role merely to run this diagnostic. Short successful `supabase_admin` sessions ending with SQLSTATE 00000 are normal disconnections, distinct from failed application connections.

Database A/B experiment

The current implementation decision is to add only the plain DISTINCT guard for exact duplicate record updates in the review code. The CASE expression that preserves an existing TOAST value remains an experimental candidate and is not adopted. No TTL normalization or production change is included. The measurements do not justify a separate `records(pk)` index without query-specific production evidence, and they quantify the database cost of removing an incorrect local quota-denial shortcut.

This is a synthetic local PostgreSQL 17.11 experiment, not a production latency or savings forecast. No production credentials, queries, DDL or payloads were used. The run created an isolated schema in a separate `query_optimization_test` database and removed that schema afterward. The active `engineering_test` application fixture was not modified.

Method and inspectable artifacts

Run `node web/tools/benchmark_postgres_queries.mjs` with `MARKET_QUERY_BENCH_DATABASE_URL` supplied privately. The script refuses non-loopback targets and database names outside `query_*_test`, including the in-use `engineering_test` database. Defaults are 256 and 100,000 synthetic rows, 41 measured trials per variant and five warmups. Scale is capped at 200,000 rows and trial count at 101. The output never includes the connection address, credentials or payload values.

Read fixtures use the actual key/value table shape: composite primary key `(pk,sk)`, partial expiry index and pending-forecast expression index. Eighty percent of rows use unique cache keys; twenty percent use multi-row journal keys. The extra-index variant adds only `(pk)`. Write comparisons use separate tables with identical indexes and data, so index differences do not confound the update guard comparison. Fixture autovacuum is disabled only inside the disposable schema; production settings are untouched.

Each variant runs in rotating/reversed order. `EXPLAIN (ANALYZE, BUFFERS, WAL, TIMING OFF, FORMAT JSON)` supplies server execution/planning time, buffer counts and WAL. Client round-trip time is recorded separately. Read cases vary their synthetic keys. A first touch after population is retained, followed by warm trials; neither PostgreSQL shared buffers nor the operating-system cache were flushed. Most measured reads were shared-buffer hits. First touch is not a disk-cold measurement. PostgreSQL EXPLAIN.

The full observations and sanitized representative plan trees are in `review/query-optimization-benchmark.json`; a compact transcript is in `review/query-optimization-benchmark.txt`. Percentiles below summarize these 41 observations, without confidence intervals. Sub-millisecond timing differences and tail values can change with desktop load and cache state.

Separate `(pk)` index

The existing composite B-tree can use its leading `pk` column. A narrower index can sometimes be cheaper, but it duplicates key storage and adds maintenance. PostgreSQL multicolumn indexes.

Rows / query	Composite only p50 / p95 ms	Plus (pk) p50 / p95 ms	Median shared-hit blocks
256 / exact (pk,sk)	0.033 / 0.047	0.031 / 0.043	3 / 3
256 / pk only	0.033 / 0.045	0.033 / 0.054	3 / 3
256 / journal prefix + ordering	0.055 / 0.071	0.056 / 0.073	16 / 16
100,000 / exact (pk,sk)	0.035 / 0.045	0.045 / 0.065	4 / 4
100,000 / pk only	0.032 / 0.058	0.032 / 0.039	4 / 4
100,000 / journal prefix + ordering	0.052 / 0.077	0.055 / 0.126	10 / 10

At 100,000 rows the additional index occupied 7,839,744 bytes, increasing total index storage from 12,697,600 to 20,537,344 bytes. Sample inserts emitted a mean 463 bytes of WAL without it versus 535 bytes with it; server insert p50/p95 was 0.031/0.042 versus 0.040/0.068 ms. The exact-key query continued to choose the composite key at scale. A pk-only tail improvement of 0.019 ms in this warm run is not evidence of an end-to-end application improvement, especially when the application already supplies sk for ordinary lookups.

Recommendation: retain the existing indexes. Reconsider only if actual pk-only query frequency, cardinality and buffers establish a material benefit. A planner's estimated ten-percent cost reduction alone does not establish measured performance or justify an extra index.

Record update comparisons

The three candidates were:

1. Existing unconditional ON CONFLICT ... DO UPDATE. 2. A WHERE payload IS DISTINCT FROM EXCLUDED.payload OR expires_at IS DISTINCT FROM EXCLUDED.expires_at guard. 3. The same guard plus SET payload=CASE WHEN payload IS NOT DISTINCT FROM EXCLUDED.payload THEN payload ELSE EXCLUDED.payload END, preserving the existing value when equal.

ON CONFLICT can lock a conflicting row even when its update predicate rejects the update; skipping the update does not mean zero locking or zero WAL. PostgreSQL INSERT.

Four payload profiles were measured: 256-character compressible padding, 16-KiB compressible padding, 16-KiB deterministic high-entropy padding and 32-KiB high-entropy padding. "Incompressible" describes these controlled high-entropy fixtures, not arbitrary production JSON. Logical JSON size includes object fields in addition to padding. Every profile covers exact duplicate, changed payload and expiry-only changes at both table sizes.

100,000 rows / 32-KiB high-entropy payload	Baseline p50 / p95 ms	DISTINCT guard p50 / p95 ms	Guard + preserve CASE p50 / p95 ms	Mean WAL bytes: baseline / guard / CASE
Exact duplicate	0.602 / 0.683	0.119 / 0.145	0.118 / 0.160	36,448 / 54 / 54
Changed payload	0.576 / 0.783	0.619 / 0.802	0.683 / 0.925	36,464 / 36,493 / 36,497
Expiry only	0.576 / 0.735	0.641 / 0.907	0.233 / 0.365	36,455 / 36,446 / 421

For exact duplicates, the baseline produced 46 updates and 46 additional dead-tuple estimates across warmups plus trials. Both guarded variants produced zero updates and retained the same tuple identity. They still emitted a 54-byte lock WAL record per measured attempt. For expiry-only changes all variants created 46 heap updates, as required; the CASE variant substantially reduced TOAST/WAL work while preserving the changed expiry. These counters are short-run fixture evidence, not a measurement of production bloat.

At 256 rows, the same 32-KiB duplicate case measured baseline p95 1.578 ms and 36,454 mean WAL bytes versus CASE p95 0.149 ms and 54 bytes. Expiry-only measured 1.245 ms / 36,458 bytes versus CASE 0.910 ms / 426 bytes. Changed-payload CASE comparisons consumed extra CPU and did not reduce WAL. For small inline payloads the CASE can also add CPU without a storage benefit. This is a workload-dependent tradeoff, not a universal speed improvement.

Recommendation: adopt only the plain DISTINCT guard now, preserving every actual payload or TTL change. The expiry-only CASE fixture deliberately changes only the SQL timestamp. Actual `save` calls embed `expires_epoch` inside JSON, and cached values often contain a new `as_of`; refreshing those fields changes the JSON, so the CASE's old-value branch cannot apply. Its synthetic expiry-only benefit is not evidence of an improvement for those application writes. CASE remains deferred unless actual callers demonstrate column-only expiry changes with equal JSON, and its extra CPU cost on changed payloads must remain part of that decision. No TTL fields are removed or normalized. The measured query uses a payload-first OR predicate; an expiry-first predicate may avoid a redundant equality comparison but was not separately timed.

Exhausted quota cost

The rejected UPSERT retains the validated `amount <= limit` insertion rule and the existing conflict guard `used <= limit-amount`. Removing a local denial cache restores database-authoritative decisions but makes each denial a database request. This experiment does not claim a missing overrun fix.

An exhausted-key UPSERT measured server p50/p95 0.056/0.072 ms, three median shared-hit blocks, zero shared reads and 54 WAL bytes from row locking. Four workers made 400 rejected requests on one hot key with round-trip p50/p95 3.581/5.052 ms. Four separate exhausted keys measured 1.971/2.850 ms. All 800 requests were denied and all five counters remained exactly 10. These local round trips include client/IPC and row-lock contention; they do not predict cross-region production latency.

This is an explicit correctness/cost tradeoff. Keep existing admission limits and the four-connection pool. A future denial cache would need authoritative expiry, exact request semantics and distributed refund/transaction invalidation; the old key-only shortcut cannot be justified by its lower latency.

Research source coverage and gap analysis

Source review completed on November 4, 2026 ET. The review covers three source inputs, their cited references, and additional primary literature. The machine-readable ledger records source identities, access outcomes, and reading scope.

Coverage and interpretation

- **161 supplied occurrences -> 97 distinct domain/title entries across 61 requested domains.** All 97 received a public open attempt or an explicit unresolved-target lookup. Two supplied titles map to arXiv 2403.09267, leaving 96 canonical URL slots; mirrors of the same paper are still separately tracked for requested-site coverage.

- **78 exact targets opened; 6 resolved targets blocked; 13 ambiguous or unresolved entries.** An open checkbox means the exact public page was reached. It does **not** mean its full contents were read or its claims were validated. Of the requested entries, 11 received selected-section reading and 9 received abstract reading; the rest are metadata/excerpt checks or incomplete targets. The duplicate arXiv entry is not counted as another substantive reading.

- **20 additional unique primary papers or official documents** were checked, across 9 canonical domains. All have an explicit reading scope below. The Hawkes 1971 publisher was unavailable, so its original published PDF was read through a public mirror. This is one paper, not two sources for the unique count.

- Three additional URLs were already cited in project documents: Gneiting-Raftery, Kalshi WebSocket sequence semantics, and PostgreSQL clock semantics. The remaining **17 canonical URLs were absent from the scanned docs/review Markdown and JSON before this ledger**. This is a reference-coverage observation, not a claim of research originality or first discovery. "Additional" below means additional to the supplied list.

- No videos were watched, gated downloads retrieved, authenticated pages bypassed, credentials accessed, paid feeds called, deployment performed or profitability/latency certified. A failed open was not retried through access evasion.

Attachment provenance

Input	Bytes	SHA-256	Reading
attachment_1	33689	`f85a71ac30e8537471623f22f4d3ade1d3007c3848306083b3994027886669bc`	Complete, bounded segments
attachment_2	5240	`629b329c825f5e7af4239c30ec475106699b50e4ebc418a9eb427c2b4db62fae`	Complete, bounded segments

Input	Bytes	SHA-256	Reading
attachment_3	19361	`81ef662dcce21063b2a7147155c5760f756e5ca74b14f0f2afd433f7b959f2b6`	Complete, bounded segments

Attachment 1 is a synthesis with equations and architectural assertions. Attachments 2-3 are repeated source titles and research notes, generally without authors or full URLs. Statements that research was being performed in those notes are not treated as independent evidence that a source was actually checked.

Corrections and application implications

F01: Kalshi matching versus LMSR

Mechanisms must be separated. Kalshi documents limit-order entry and complementary YES/NO bid ladders. LMSR is a cost-function market maker with its own state and liquidity parameter.

Keep LMSR as an explicitly illustrative mechanism. Paper fills must consume actual order-book depth and current fee/rounding rules; LMSR marginal prices are not Kalshi fills. Primary source 1, Primary source 2, Primary source 3.

F02: Reservation-price direction

Reversed trading intuition. In A-S, r=s-qgammasigma^2(T-t). Positive inventory lowers reservation value to favor selling. With other conditions fixed, a lower bid discourages sellers hitting it, while a lower ask attracts buyers lifting it.

Test sign symmetry and units. Do not apply an unbounded Brownian stock-price model to binary probabilities without boundary/settlement treatment. Read scope: Reservation definitions and Eq. 2.6-2.7 / 3.17, PDF pp3 and 9. Primary source.

F03: GLFT bid sign and HJB transform

Bid sign error plus an overbroad methods claim. Theorem 1 defines bid offset deltab=s-bid and ask offset deltaa=ask-s. Both include c=log(1+gamma/k)/gamma. When adjacent v values coincide, bid=s-c and ask=s+c. The paper explicitly transforms its HJB system into linear ODEs.

Derive prices from offsets consistently. Separate finite-horizon solutions from asymptotic approximations and do not promise nanosecond runtime. Read scope: Abstract and Theorem 1, PDF p6. Primary source.

F04: Classic OFI, queue imbalance and conflation

Incorrect data requirement and missing transport condition. Cont et al. use changes in best bid/ask prices and queue sizes (Level 1). MLOFI extends across levels. Static depth imbalance is different. Kalshi may net per-price changes during conflation while keeping seq contiguous, so exact event ordering can still be lost.

Label observed-stream estimates with sampling/conflation limitations. Maintain bounded pre-throttle event windows only under stated transport assumptions; clear on snapshots/gaps. Contemporaneous explanation is not automatically future-return prediction. Read scope: Cont data section p7; Kalshi conflation and gap-recovery rules. Primary source 1, Primary source 2, Primary source 3.

F05: Almgren-Chriss discreteness and finite risk

Continuous-limit approximation was presented as exact; variance claim is overstated. The discrete paper defines 2[cosh(kappatau)-1]/tau^2=kappa?^2 and obtains the square-root expression as tau tends to zero. Its optimal path solves expected execution cost plus lambda times variance under specified impact/price assumptions; it does not establish a directional entry signal.

Show the continuous approximation explicitly or use the discrete relation. Retain finite variance and impact assumptions; realized cost estimates need actual depth, fees and fill uncertainty. Read scope: Expected cost/variance and efficient frontier; Eq. 17-19, PDF pp10-14. Primary source.

F06: DeepLOB and GPU/inference claims

Conflates distinct studies and training with inference. DeepLOB uses ten price/volume levels and reports training on one Tesla P100; a different 2016 spatial-network paper reports 50 GPUs across nearly 500 stocks. The digital-asset paper reports commodity-GPU training and a two-second forecast horizon. Those are not app inference p99 measurements.

Benchmark actual model/hardware/batch/cold/warm runtime and memory. DeepLOB's simplified gross-before-fee exercise does not establish live profitability. Read scope: DeepLOB data and experimental-setup sections; other papers' abstracts. Primary source 1, Primary source 2, Primary source 3.

F07: Forecast quality versus tradability

Unsupported inference. The requested microstructural forecasting guide explicitly separates machine-learning metrics from complete-transaction practicability and calls for cross-exchange validation. Proper scores evaluate probabilities; trial selection can inflate backtest statistics.

Keep chronological splits, overlapping-label precautions, baselines, calibration, trial records, fees, slippage, queue/fill assumptions and confidence intervals. None of these papers certifies the application's profit or novelty. Primary source 1, Primary source 2, Primary source 3, Primary source 4.

F08: Hawkes feasibility and stability

Overbroad computational and mathematical claims. Exponential-kernel likelihood/derivatives admit recursive evaluation. Required performance depends on event rate, dimension, optimizer and update cadence. Nonnegative linear Hawkes stationarity conditions do not transfer unchanged to every inhibitory/nonlinear variant.

Fit and benchmark bounded models with residual/held-out checks. Do not treat fitted event intensity as a calibrated contract-resolution probability. Read scope: Laub estimation/branching sections; Bonnet et al. abstract; Bacry survey Appendix C.1 (quant agent read). Primary source 1, Primary source 2, Primary source 3.

F09: CPU isolation, cache lines and locks

Multiple systems overstatements. Linux documents housekeeping and workload constraints; the C++ draft defines implementation-dependent constexpr interference sizes. Layout and access patterns matter for false sharing. Linux futex fast paths can stay in userspace. The Disruptor's default wait strategy explicitly uses locks/conditions.

Use workload-specific profiling and latency distributions. NIC/kernel tuning is not a demonstrated optimization for a hosted Node/REST app, and busy polling spends CPU. Primary source 1, Primary source 2, Primary source 3, Primary source 4, Primary source 5, Primary source 6.

F10: LMSR scope and loss bound

Requires explicit outcome/state assumptions. For one mutually exclusive exhaustive n-outcome market with zero initial inventory, C(q)=b log sumexp(qi/b) implies max_i qi-C(q)+C(0)<=b log(n). This is a derivation under that payout model, not a claim about all linked contracts, fees, or Kalshi.

Define outcome sets, initial state and settlement compatibility. Separate local price normalization from cross-contract no-arbitrage and operational risk. Mathematical inference from the stated cost/payout assumptions. Primary source.

These corrections do not prove that other HJB/Feynman-Kac formulations are impossible or universally interchangeable. No viscosity-solution proof or exhaustive stochastic-control treatment was completed in this source lane. The specific GLFT transformation is sufficient to reject the paper's categorical impossibility claim. Likewise, market-making inventory control, optimal liquidation and event-outcome probability prediction have different objectives; combining their names does not establish a unified profitable strategy.

Requested source checklist

`[x]` = exact target opened, with reading scope stated. `[ ]` = blocked or unresolved/ambiguous. Public-site or near-title candidates for unchecked entries do not count as successful target retrieval. Secondary pages are discovery/context only unless explicitly indicated; the corrections above rely on original papers or official documentation.

- [x] **R001 - reddit.com: r/quant on Reddit: Solarflare ef_vi vs Onload benchmarked on** (source). **opened; metadata/excerpt.** One self-reported Solarflare X2522-Plus benchmark; hardware and traffic details prevent treating its timing table as a universal latency guarantee.

- [x] **R002 - docs.amd.com: ef_vi - UG1586** (source). **opened; selected sections.** Official Onload UG1586 appendix for ef_vi; vendor scope is compatible adapters and layer-2 access, not cloud HTTP latency.

- [x] **R003 - ramcloud.atlassian.net: View Source** (source). **opened; metadata/excerpt.** RAMCloud engineering notes resolve the generic 'View Source' title to ef_vi ring-buffer setup; project notes are not universal benchmark evidence.

- [x] **R004 - amd.com: ef_vi User Guide \| AMD** (source). **opened; metadata/excerpt.** Resolved AMD's guide through its official download index; current guide is SF-114063-CD revision 20, distinct from Onload UG1586.

- [x] **R005 - github.com: The latency of ef_eventq_poll() and ef_vi_transmit() is not stable. #48** (source). **opened; metadata/excerpt.** Issue #48 is a user report of unstable polling/transmit timings, useful as a counterexample to deterministic-latency claims, not a controlled benchmark.

- [x] **R006 - docs.kernel.org: CPU Isolation - The Linux Kernel documentation** (source). **opened; selected sections.** CPU isolation requires housekeeping/offloading and workload restrictions; it does not prohibit all kernel work or interruptions.

- [x] **R007 - docs.kernel.org: The kernel's command-line parameters** (source). **opened; selected sections.** Read nohz_full/isolcpus portions, not all 8,630 lines; flags have configuration and workload conditions.

- [x] **R008 - access.redhat.com: An overview and comparison of the isolcpus kernel parameter** (source). **opened; metadata/excerpt.** Customer-portal solution opened; subscriber-only material is not treated as read or bypassed.

- [x] **R009 - nimrod.blog: A Subtle Bug Before C++17 \| Nimrod's Coding Lab** (source). **opened; metadata/excerpt.** Metadata/excerpt addresses pre-C++17 allocation of over-aligned types; implementation details must be checked against the language version.

- [x] **R010 - ryonaldteofilo.medium.com: Cache Line Alignment in C++ - How It Makes Your Program Faster.** (source). **opened; metadata/excerpt.** Blog article opened; cache-alignment discussion is secondary to the C++ draft and measured structure layout.

- [ ] **R011 - medium.com: Data Alignment in C++ - What It Is, Why It Hurts When Ignored, and** (site/candidate). **unresolved/ambiguous; target not read.** Exact article/author URL unresolved. A Medium curated-list discovery page opened but exposed no target article content.

- [x] **R012 - reddit.com: Idiomatic way to avoid false sharing : r/cpp - Reddit** (source). **opened; metadata/excerpt.** Public r/cpp thread opened; discussion provides examples, not a language-standard guarantee.

- [x] **R013 - moderncpp.dev: Cache-Line Archaeology: Finding and Fixing False Sharing in** (source). **opened; metadata/excerpt.** Secondary article opened; its own CPU benchmark cannot establish a universal 64-byte cache line or app-wide speedup.

- [x] **R014 - stackoverflow.com: How does LMAX's disruptor pattern work? - Stack Overflow** (source). **opened; metadata/excerpt.** Discussion page opened; LMAX's official guide is authoritative for selectable producers/wait strategies.

- [x] **R015 - medium.com: Pattern for concurrency in latency-sensitive systems \| by Aniket Bidgar** (source). **opened; metadata/excerpt.** Aniket Bidgar article resolved and opened; implementation commentary is secondary, not proof of end-to-end latency.

- [x] **R016 - sanjeev.pages.dev: The LMAX Disruptor Codebase \| Stay Tuned - Sanjeev Sarda** (source). **opened; metadata/excerpt.** Codebase walkthrough opened; not used as authority for profit or portable performance.

- [x] **R017 - lmax-exchange.github.io: LMAX Disruptor User Guide** (source). **opened; selected sections.** Read wait-strategy section: default BlockingWaitStrategy uses a lock/condition variable;

claiming all Disruptor configurations avoid locks is incorrect.

- [ ] **R018 - medium.com: From Wall Street to Code: The Performance Secrets of LMAX - Medium** (site/candidate). **unresolved/ambiguous; target not read.** Exact Medium article unresolved despite exact-title searches. Medium homepage opened; similarly titled Substack index is only a discovery candidate.

- [x] **R019 - semicolony.dev: How Ring Buffers Work: Single-Producer, Multi-Consumer, Lock-Free** (source). **opened; metadata/excerpt.** Interactive explainer opened; producer/consumer model and memory ordering still need implementation-specific checks.

- [ ] **R020 - ora.ox.ac.uk: Multi-Level Order-Flow Imbalance in a Limit Order Book** (site/candidate). **blocked; target not read.** Oxford repository file returned 403. The same paper's arXiv abstract was read through an alternate public host. Public alternative.

- [x] **R021 - microalphas.com: Order Flow Imbalance (OFI): Reading Short-Horizon Price Pressure** (source). **opened; metadata/excerpt.** Secondary OFI explainer opened; classical formula and required inputs were verified in Cont et al.'s original paper.

- [x] **R022 - scribd.com: Price Impact of Order Flow Imbalance \| PDF \| Econometrics - Scribd** (source). **opened; metadata/excerpt.** Scribd page metadata opened; no gated download attempted. Same intellectual work as Cont et al. arXiv 1011.6402. Public alternative.

- [x] **R023 - arxiv.org: Maximum Likelihood Estimation for Hawkes Processes with self** (source). **opened; abstract.** Abstract read: univariate likelihood method explicitly handles excitation or inhibition; do not generalize nonnegative-kernel stationarity rules to every variant.

- [x] **R024 - spiral.imperial.ac.uk: Parameter Estimation of Binned Hawkes Processes - Spiral** (source). **opened; metadata/excerpt.** Imperial accepted-paper PDF opened; title/abstract excerpt checked. Binned-count estimation differs from exact event-time likelihood.

- [ ] **R025 - tandfonline.com: Full article: Parameter Estimation of Binned Hawkes Processes** (site/candidate). **blocked; target not read.** Publisher returned 403. Accepted-manuscript alternative R024 is public; publisher full text was not read.

- [ ] **R026 - mdpi.com: Quantitative and Comparative Analyses of Limit Order Books ... - MDPI** (site/candidate). **blocked; target not read.** Publisher returned 429; stopped. Search metadata resolves He and Svischuk's 2019 compound-Hawkes LOB paper; full text not read.

- [ ] **R027 - paperswithbacktest.com: Hawkes Process in Trading: Self-Exciting Models of Trade Clustering** (site/candidate). **unresolved/ambiguous; target not read.** Exact Hawkes article URL unresolved. Site homepage opened; a guessed wiki route also failed and is not represented as a resolved source.

- [x] **R028 - lup.lub.lu.se: PARAMETER ESTIMATION FOR HAWKE'S PROCESSES** (source). **opened; metadata/excerpt.** Lund student-paper record opened; student thesis metadata is not peer-reviewed market-performance evidence.

- [x] **R029 - scribd.com: DeepLOB: CNNs for Limit Order Books \| PDF - Scribd** (source). **opened; metadata/excerpt.** Scribd metadata opened; original DeepLOB arXiv paper used for technical checks.

- [x] **R030 - arxiv.org: DeepLOB: Deep Convolutional Neural Networks for Limit Order Books** (source). **opened; selected sections.** Read abstract and selected original-PDF sections: ten price/volume levels, single P100 training, and simplified gross-before-fee experiments. Public alternative.

- [ ] **R031 - scholar.google.com: Zihao Zhang - Google Scholar** (site/candidate). **unresolved/ambiguous; target not read.** Exact Zihao Zhang Scholar profile unresolved. Scholar homepage reports inability to perform operation; author's publication page is an alternate discovery source. Alternative candidate.

- [x] **R032 - arxiv.org: Deep Limit Order Book Forecasting A microstructural guide - arXiv** (source). **opened; selected sections.** Read abstract and selected HTML conclusion; high classification performance need not yield actionable transactions; cross-exchange validation remains necessary. Public alternative.

- [x] **R033 - arxiv.org: [1601.01987] Deep Learning for Limit Order Books - arXiv** (source). **opened; abstract.** Abstract read. This separate 2016 spatial-network paper reports a 50-GPU training cluster; do not attribute that configuration to DeepLOB.

- [x] **R034 - maths.ox.ac.uk: Algorithmic market making for options** (source). **opened; metadata/excerpt.** Author's Oxford options-market-making slide deck opened; metadata/intro checked, not all 83 slides.

- [x] **R035 - stanford.edu: Optimal High-Frequency Market Making - Stanford University** (source). **opened; metadata/excerpt.** Stanford course-project presentation opened; student simulation results are not a general theorem or independently replicated trading result.

- [x] **R036 - people.orie.cornell.edu: High-frequency trading in a limit order book** (source). **opened; selected sections.** Read reservation-price definitions and approximation sections. Positive inventory lowers reservation value to encourage liquidation. Public alternative.

- [x] **R037 - riskhub.org: Hamilton-Jacobi-Bellman Equations Explained - Risk Hub** (source). **opened; metadata/excerpt.** Secondary HJB explainer opened; it cannot justify blanket impossibility claims about stochastic-control methods.

- [x] **R038 - arxiv.org: High-frequency market-making with inventory constraints and ... - arXiv** (source). **opened; abstract.** Abstract read: extends inventory-constrained market making to directional mid-price dynamics under stated utility assumptions.

- [x] **R039 - hftbacktest.readthedocs.io: Gue?ant-Lehalle-Fernandez-Tapia Market Making Model - HftBacktest** (source). **opened; metadata/excerpt.** Official HftBacktest tutorial opened; library-specific approximation/calibration instructions require their own simulation assumptions.

- [x] **R040 - researchgate.net: Dealing with the Inventory Risk: A Solution to the Market Making** (source). **opened; metadata/excerpt.** ResearchGate metadata/abstract opened; original GLFT arXiv used instead of the mirror for mathematical verification.

- [x] **R041 - api.pageplace.de: The Financial Mathematics of Market Liquidity** (source). **opened; metadata/excerpt.** Publisher book preview opened; only preview portions are available, not the entire book.

- [x] **R042 - quantmemo.com: Gueant, Lehalle and Fernandez-Tapia on Inventory Risk** (source). **opened; metadata/excerpt.** Secondary inventory-risk article opened; GLFT original governs formula signs and assumptions.

- [x] **R043 - medium.com: Deep Dive into IS: The Almgren-Chriss Framework \| by Anboto Labs** (source). **opened; metadata/excerpt.** Anboto Labs article opened; implementation/marketing claims are not independent validation of execution savings.

- [x] **R044 - metricgate.com: Almgren-Chriss Optimal Execution Calculator - MetricGate** (source). **opened; metadata/excerpt.** Calculator documentation opened; parameter units, discrete/continuous formulas, and transaction-cost assumptions need explicit validation.

- [x] **R045 - quantt.co.uk: Almgren-Chriss Optimal Execution Explained (2026) - Quantt** (source). **opened; metadata/excerpt.** Secondary 2026 guide opened; publication freshness alone does not validate model calibration.

- [x] **R046 - quantum-blog.medium.com: Almgren-Chriss Optimal Execution Model - Quantum - Medium** (source). **opened; metadata/excerpt.** Secondary model article opened; use the original Almgren-Chriss paper for the mathematical result.

- [ ] **R047 - imperial.ac.uk: Market impact models and optimal execution algorithms** (site/candidate). **blocked; target not read.** Imperial Lecture1 PDF returned 403; search metadata resolves lecture sequence. Full slide content not read.

- [x] **R048 - dm13450.github.io: Solving the Almgren Chris Model - Dean Markwick** (source). **opened; metadata/excerpt.** Implementation walkthrough opened; not independently executed or numerically audited in this source-only lane.

- [x] **R049 - gensyn.ai: LMSR (Logarithmic Market Scoring Rule) - Gensyn** (source). **opened; metadata/excerpt.** Gensyn's LMSR guide opened; describes that mechanism, not evidence Kalshi uses it.

- [x] **R050 - cultivatelabs.com: How does the Logarithmic Market Scoring Rule (LMSR) work?** (source). **opened; metadata/excerpt.** Vendor LMSR explanation opened; original Hanson's paper supplies primary mathematical provenance.

- [x] **R051 - predictionmarketmath.org: Logarithmic Market Scoring Rules: Hanson's LMSR Cost Function** (source). **opened; metadata/excerpt.** Exact article found through English homepage link and opened. Site's self-described peer-review/team credentials were not independently verified.

- [x] **R052 - github.com: Build Interactive LMSR (Logarithmic Market Scoring Rule ... - GitHub** (source). **opened; metadata/excerpt.** GitHub issue is a pedagogical-demo request, not an original LMSR derivation or verified production implementation.

- [ ] **R053 - community.nasscom.in: How Prediction Market Algorithms Work: LMSR Explained** (site/candidate). **unresolved/ambiguous; target not read.** Exact community article unresolved; community homepage returned 406. Dailyhunt syndication title appeared in discovery, not read as primary evidence.

- [ ] **R054 - researchgate.net: LOGARITHMIC MARKETS CORING RULES FOR MODULAR** (site/candidate). **unresolved/ambiguous; target not read.** Exact ResearchGate record unresolved. Homepage is a login page; original Hanson author-hosted paper is available at R083. Public alternative 1, Public alternative 2.

- [x] **R055 - quant.stackexchange.com: Market Making Strategies Found by Hamilton-Jacobi-Bellman** (source). **opened; metadata/excerpt.** Quant Stack Exchange question resolved and opened; discussion is not primary proof of HJB correctness.

- [x] **R056 - ideas.repec.org: Macroscopic Market Making - IDEAS/RePEc** (source). **opened; metadata/excerpt.** RePEc metadata/abstract opened; routes to Guo, Jin and Nam's Macroscopic Market Making, arXiv 2307.14129.

- [ ] **R057 - youtube.com: The Avellaneda-Stoikov Market Making Model - YouTube** (site/candidate). **unresolved/ambiguous; target not read.** One plausible exact-titled YouTube video identified; open was throttled. No video/audio/transcript watched or accuracy certification made.

- [x] **R058 - web.stanford.edu: Stochastic Control for Optimal Market-Making - Stanford University** (source). **opened; metadata/excerpt.** Ashwin Rao's Stanford teaching slides opened; notation/intro checked, not a peer-reviewed model-performance study.

- [x] **R059 - arxiv.org: [2403.09267] Deep Limit Order Book Forecasting - arXiv** (source). **opened; metadata/excerpt.** Canonical duplicate of R032 (arXiv 2403.09267); supplied short and subtitle-bearing titles refer to the same paper.

- [x] **R060 - ideas.repec.org: DeepLOB: Deep Convolutional Neural Networks for Limit Order Books** (source). **opened; metadata/excerpt.** RePEc metadata opened; same intellectual work as DeepLOB R030.

- [x] **R061 - arxiv.org: Market Making with Deep Reinforcement Learning from Limit Order** (source). **opened; abstract.** Abstract read: Attn-LOB reinforcement-learning agent, continuous actions and hybrid reward; no reproduction or live profitability established.

- [x] **R062 - quantmemo.com: Optimal Execution, Explained - Quant Memo** (source). **opened; metadata/excerpt.** Secondary optimal-execution overview opened; does not validate a fill model or projected savings.

- [x] **R063 - arxiv.org: Optimal order execution subject to reference strategies - arXiv** (source). **opened; abstract.** Abstract read: client reference strategies under execution risk; closed forms depend on the stated broker-dealer utility objective.

- [x] **R064 - scribd.com: Almgren - Chriss - Optimal Execution \| PDF \| Market (Economics)** (source). **opened; metadata/excerpt.** Scribd metadata opened; no gated download or authenticated retrieval attempted.

- [x] **R065 - github.com: Optimal-Portfolio-Transactions/Almgren and Chriss Model.ipynb at** (source). **opened; metadata/excerpt.** Notebook page opened; metadata checked, notebook not executed or accepted as calibrated trading evidence.

- [x] **R066 - arthur.bagourd.com: A Tale of Two Models: Implementing the Almgren-Chriss framework** (source). **opened; metadata/excerpt.** Author's 29-page implementation report opened; title/abstract excerpt checked, not the full report or notebooks.

- [x] **R067 - smallake.kr: Optimal Execution of Portfolio Transactions?** (source). **opened; selected sections.** Original Almgren-Chriss paper hosted by a third party; selected sections on expected cost, variance, optimal path and discrete kappa read.

- [ ] **R068 - researchgate.net: FPGA Based Low-Latency Market Data Feed Handler - ResearchGate** (site/candidate). **unresolved/ambiguous; target not read.** Exact requested ResearchGate work unresolved. Opened near-title Allred feed-handler page is a different work; Zhou et al.'s matching title found in Springer book contents. Alternative candidate.

- [x] **R069 - scribd.com: 3-Optimal Market Making - Applied Mathematical Finance - Scribd** (source). **opened; metadata/excerpt.** Scribd page metadata opened; Gue?ant's Optimal Market Making paper is the primary alternative, arXiv 1605.01862. Public alternative.

- [x] **R070 - arxiv.org: [1105.3115] Dealing with the Inventory Risk. A solution to the market** (source). **opened; selected sections.** Read original abstract and Theorem 1: linear ODE transform and bid/ask offset signs. Closed-form approximations are not portable nanosecond guarantees. Public alternative.

- [x] **R071 - arxiv.org: Closed-form approximations in multi-asset market making - arXiv** (source). **opened; abstract.** Abstract read: multi-asset approximations/proxies, finite-horizon and ergodic regimes; not exact universal solutions.

- [x] **R072 - ideas.repec.org: Dealing with the Inventory Risk. A solution to the market making** (source). **opened; metadata/excerpt.** RePEc metadata opened; same intellectual work as GLFT R070.

- [x] **R073 - emergentmind.com: Order-Flow Imbalance (OFI) - Emergent Mind** (source). **opened; metadata/excerpt.** Aggregator OFI article opened; not a substitute for original formula, sampling requirements or held-out evaluation.

- [x] **R074 - scribd.com: Solarflare Ef Vi User Guide \| PDF - Scribd** (source). **opened; metadata/excerpt.** Scribd metadata opened; current AMD guide used for authoritative API provenance.

- [x] **R075 - github.com: openonload/src/include/etherfabric/ef_vi.h at master - GitHub** (source). **opened; metadata/excerpt.** Historical majek OpenOnload fork header opened; fork age/API compatibility matters. Not the current vendor-support contract.

- [x] **R076 - redhawk.concurrent-rt.com: Onload User Guide - Concurrent Real-Time Software Portal** (source). **opened; metadata/excerpt.** Public mirror of Onload UG1586 v1.2 July 2023 opened; not current revision and not all 424 pages read.

- [ ] **R077 - news.ycombinator.com: It's an absolutely wonderful article - perhaps one of the best I've** (site/candidate). **unresolved/ambiguous; target not read.** Exact HN comment unresolved despite distinctive-quote searches. HN homepage opened; Algolia public-search attempt was inaccessible.

- [ ] **R078 - adaptivesupport.amd.com: multicast packet loss in ef_vi processes while onload processes are** (site/candidate). **unresolved/ambiguous; target not read.** Exact AMD support thread

unresolved. Public support homepage renders Loading/CSS Error; no login or workaround attempted.

- [x] **R079 - blog.hieunt.me: LMAX Disruptor: A Superior Alternative to Bounded Queues for** (source). **opened; metadata/excerpt.** Secondary Disruptor walkthrough opened; cannot replace official wait-strategy semantics.

- [x] **R080 - lmax-exchange.github.io: LMAX Disruptor: High performance alternative to bounded queues** (source). **opened; metadata/excerpt.** Official project landing page opened; library throughput claims remain workload-dependent.

- [x] **R081 - martinfowler.com: The LMAX Architecture - Martin Fowler** (source). **opened; metadata/excerpt.** Fowler architecture article opened; architecture exposition is not a promise of equivalent Node/cloud performance.

- [ ] **R082 - youtube.com: LMAX Disruptor and the Concepts of Mechanical Sympathy - YouTube** (site/candidate). **unresolved/ambiguous; No video/audio/transcript consumed.** Exact YouTube video URL unresolved. Matching Jamie Allen Vimeo page found; video/transcript was not watched/read. Alternative candidate.

- [x] **R083 - mason.gmu.edu: Logarithmic Market Scoring Rules for Modular Combinatorial** (source). **opened; selected sections.** Author-hosted Hanson paper opened; selected cost/scoring-rule discussion checked. Discrete finite-outcome LMSR needs its own assumptions.

- [x] **R084 - mathoverflow.net: Asymptotic bounds on the maximum loss of Logarithmic Market** (source). **opened; metadata/excerpt.** Exact canonical MathOverflow question opened after correcting discovery slug; continuous-state discussion is distinct from finite-outcome b log(n).

- [x] **R085 - docs.kalshi.com: Order Entry - Kalshi's API Documentation** (source). **opened; selected sections.** Read FIX new-order section: limit orders, post-only option, quantity/price conventions. This is exchange order entry, not an LMSR pricing specification.

- [x] **R086 - stackoverflow.com: Aligning to cache line and knowing the cache line size** (source). **opened; metadata/excerpt.** Discussion page opened; platform cache-line detection and alignment are not universally equivalent.

- [x] **R087 - stackoverflow.com: False sharing prevention with alignas is broken - Stack Overflow** (source). **opened; metadata/excerpt.** Discussion page opened; aligning a containing structure does not automatically separate every independently written field.

- [x] **R088 - arxiv.org: Optimal Execution of Limit and Market Orders with Trade ... - arXiv** (source). **opened; abstract.** Abstract read: permanent/temporary impact plus fill uncertainty, terminal penalty and parameter conditions; not guaranteed limit fills.

- [ ] **R089 - tandfonline.com: Full article: Optimal Execution: A Review** (site/candidate). **blocked; target not read.** Publisher full text returned 403; metadata/abstract discoverable only. No paywall bypass.

- [ ] **R090 - resolve.cambridge.org: Optimal Execution - Cambridge University Press & Assessment** (site/candidate). **unresolved/ambiguous; target not read.** Supplied generic resolve.cambridge title is ambiguous and root returned 403. Public Cambridge chapter candidate

opened; exact intended chapter remains unconfirmed. Alternative candidate.

- [ ] **R091 - math-evry.cnrs.fr: Optimal execution cost for liquidation through a limit order market** (site/candidate). **blocked; target not read.** Supplied CNRS domain's title resolved to an Evry author-repository PDF, but open timed out. No full-text read.

- [ ] **R092 - researchgate.net: Optimal execution with limit and market orders \| Request PDF** (site/candidate). **unresolved/ambiguous; target not read.** ResearchGate metadata opened for an exact-title candidate; more than one paper has this generic title, so intended identity remains ambiguous.

- [x] **R093 - github.com: Almgren-Chriss - DYNAMIC PROGRAMMING.ipynb - GitHub** (source). **opened; metadata/excerpt.** Notebook page opened; not executed, audited or claimed to match the optimal strategy.

- [x] **R094 - mathematicsconsultants.com: Optimal Liquidation Algorithms - the Almgren-Chriss Model** (source). **opened; metadata/excerpt.** Secondary consultant explainer opened; not evidence of realized client savings or calibrated app behavior.

- [x] **R095 - simtrade.fr: Understanding the Almgren-Chriss Model for Optimal Trade Execution** (source). **opened; metadata/excerpt.** Exact SimTrade article resolved and opened; secondary pedagogy, not original mathematical proof.

- [x] **R096 - arxiv.org: Optimal Execution with Dynamic Order Flow Imbalance - arXiv** (source). **opened; abstract.** Abstract read: execution control includes informational and instantaneous impact, with approximate receding-horizon solutions.

- [x] **R097 - arxiv.org: Deep Learning for Digital Asset Limit Order Books - arXiv** (source). **opened; abstract.** Abstract read: commodity-GPU training, two-second Bitcoin forecast horizon and reported walk-forward accuracy are market/study-specific.

Twenty additional primary sources

These are distinct papers or documents, not alternate hosts counted twice. Abstract-only reads support the described study scope; they do not imply independent replication or a full theorem audit. The original Cont OFI paper, authoritative alternatives for Scribd/ResearchGate items, and blocked Brier paper are excluded from this count.

- [x] **A001: Queue Imbalance as a One-Tick-Ahead Price Predictor in a Limit Order Book** - abstract_read. Paper abstract; logistic-regression prediction with stock-dependent results, not executable P&L.

- [x] **A002: Price dynamics in a Markovian limit order market** - abstract_read. Stylized best-queue model; transition assumptions require calibration.

- [x] **A003: A Correction Note for Price Dynamics in a Markovian Limit Order Market** - Publisher correction abstract. Published correction must accompany original model; do not silently rely on old formulas.

- [x] **A004: Simulating and analyzing order book data: The queue-reactive model** - abstract_read. Queue-state-dependent order intensities; simulator assumptions and execution probabilities require

market-specific validation.

- [x] **A005: Spectra of some self-exciting and mutually exciting point processes** - Published-paper summary/model introduction via public mirror. Original Hawkes paper; covariance and excitation theory, not a profitability theorem. Public alternative.

- [x] **A006: Hawkes processes in finance** - abstract_read. Author survey; multivariate excitation, kernels, estimation and empirical limits.

- [x] **A007: Hawkes Processes** - Abstract, branching-ratio and recursive likelihood sections. Author review; simulation, likelihood estimation and validation need event timestamps. Public alternative.

- [x] **A008: Hawkes Models And Their Applications** - abstract_read. Updated author survey; estimation and application assumptions differ across domains.

- [x] **A009: Strictly Proper Scoring Rules, Prediction, and Estimation** - selected_sections_read. Proper scores encourage honest probability distributions; retain outcomes and compare against baselines without claiming classification accuracy is calibration. Previously referenced in `web/docs/nba-challenger-validation.json`, `web/docs/NBA_RESEARCH.md`.

- [x] **A010: The probability of backtest overfitting** - selected_sections_read. Research-selection bias requires recorded trials and validation; no universal safe backtest count.

- [x] **A011: The Deflated Sharpe Ratio** - selected_sections_read. Selection and non-normality adjustment relies on sample and trial assumptions; not proof of future profit.

- [x] **A012: Kalshi Orderbook Responses** - selected_sections_read. YES/NO bid ladders imply complementary asks; static depth imbalance is not event-level OFI.

- [x] **A013: Kalshi Orderbook Updates** - Sequence, snapshot/gap recovery and conflation rules. Apply deltas in subscription seq order; replace on snapshot or gap. Conflation nets changes and loses individual events even with contiguous seq. Public alternative. Previously referenced in `web/docs/RUNTIME_RESEARCH.md`.

- [x] **A014: Kalshi Rate Limits and Tiers** - selected_sections_read. Token budgets and 429 backoff must be read from current docs, not assumed legacy fixed request counts.

- [x] **A015: Kalshi Fee Rounding** - selected_sections_read. Paper execution must match current rounding and fee units.

- [x] **A016: C++ draft hardware interference sizes** - selected_sections_read. Implementation-defined constexpr constants are neither universal 64 bytes nor runtime cache-line discovery.

- [x] **A017: Linux Kernel False Sharing** - selected_sections_read. Profile actual shared cache lines and structure layouts before padding.

- [x] **A018: Linux Kernel Robust Futexes** - selected_sections_read. Uncontended lock fast paths can remain in userspace; blocking contention may enter the kernel.

- [x] **A019: PostgreSQL 17 Date/Time Functions** - selected_sections_read. clock_timestamp changes during a statement; now/current_timestamp use transaction start. Previously referenced in `docs/postgres-query-hardening.md`.

- [x] **A020: PostgreSQL 17 Explicit Locking** - selected_sections_read. Locks and transaction ordering matter for quota and lease publication; isolation is not latency determinism.

Remaining limits and verification

Exact-target incompleteness is retained above rather than filled with lookalike links. In particular, the FPGA ResearchGate near-title is a different work; the generic Cambridge and ResearchGate execution titles lack enough bibliographic detail to certify their intended identity. Dynamic AMD support and Scholar pages supplied no usable target content. Publisher 403/429 responses and video throttling remain genuine retrieval limits.

The source manifest was checked for complete attachment digests, unique identifiers, all 97 requested entries, valid public HTTPS URL shapes, truthful checkbox/scope consistency, canonical duplicate handling and 20 additional unique URLs. Every quotation is short and attributed in the JSON; the ledger otherwise paraphrases. Source-only documentation changes do not require app tests. There is no repository pre-commit configuration at this review point. Other agents own the concurrent implementation and runtime validation.

Input hashes and regeneration

Install the pinned dependencies from web/tools/report-requirements.txt, then run python web/tools/build_execution_audit.py from the repository. Use --output and --public-copy to select output paths.

The hashes cover the source documents, code, and numerical fixtures with LF line endings. See the separate production verification for deployment evidence.

Input	SHA-256
review/execution-research-validation.json	e0bc8d4ce9c3075309f004d3794bb85e9100869d37b81ed12489b0ff7c836fae
review/quant-research-sources.json	a208c2843013f61a3ff9d1d356ce61557a90a6e15bd7116b60d55b7e2c453a37
docs/IMPLEMENTATION_VERIFICATION.md	13267e14009c5480fec4060e6154d7a0269ecb5b220f6c424bbf1237cc25ff09
docs/EXECUTION_RESEARCH.md	fcfda64b965e8d0355c66c490d2071173462c776da395793705a7731b73c9a12
docs/postgres-query-hardening.md	b3192419b2a3b19302f4d859131d75abb84232ac76c9e06a55fda17baa22ecdf
docs/query-optimization-benchmark.md	aa1cb15e3d9b3e621be5f96626b9839f403234aa29c1edc979cdbda30528be4f
docs/quant-research-source-ledger.md	9fdfb859090f3fb488502723af59d78716a2a659822d02f0d2deef4cacece927
web/lib/workspace/execution-research.mjs	8ff13511ac7b377a1595fd42dad44dc977c76f04d0f62137d669072622448d5c
web/lib/workspace/microstructure-model.mjs	db0a8071482368d174334acb3feba0fa9e41b9a3acec42678d877acd8aeb6d48
web/lib/workspace/execution-service.mjs	8a34ccc98e23a1dfecbc65cb4c50888ad541b470127c41895b31febec6d97d79
web/lib/cloud/microstructure-window.mjs	7009da8661eb9bcde7d48660854b8312869a33f547922075fcb172ab1e2ffe03
web/lib/cloud/feed.mjs	b9373b76725ad99f855384963b8fbb766e98d971dff61da4287ff5100edd79a3
web/tools/validate_execution_research.mjs	9514aca43b7160d38594df76489fbfd56477ec744203fd929a60bd651ac7acba

Input	SHA-256
web/tools/build_execution_audit.py	5b2be1693be2d7f6e57a76aad4711cbf6f04133a5df4a43bc6f642acb81ae365
web/tools/build_research_audit.py	52418f30be898aacc3300514e06731364fccb32747226ee685838c1e6e98e5a1
web/tools/report-requirements.txt	054e7483ed0899a13425d3ff89d0ae162290c88989825c77d22174ceb36f33f1